Day 6 Study Notes

Wed 18 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Atoms (75 min) · Maths: Matrices (75 min) · Chemistry: Periodicity (60 min)
Exam	1 h 15 min	The Day 6 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Atoms

About one question per paper, usually from the Bohr model: radius, energy, or a spectral line. Four proportionalities and the number 13.6 answer almost all of them.

1. Rutherford's alpha-scattering experiment

- Most alpha particles pass straight through thin gold foil, a few are deflected through large angles, and about 1 in 8000 bounces back. So the positive charge and nearly all the mass sit in a tiny nucleus.
- **Distance of closest approach** for a head-on alpha particle of kinetic energy K: all the kinetic energy becomes potential energy, so $r_0 = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2Ze^2}{K}$. It is inversely proportional to K.
- **Impact parameter** b is the perpendicular distance of the initial path from the nucleus. A smaller b gives a larger scattering angle; $b = 0$ gives 180°.

2. Bohr's postulates

- Electrons move in certain stable orbits without radiating.
- Angular momentum is quantised: $L = mvr = \dfrac{nh}{2\pi}$.
- A photon is emitted or absorbed when the electron jumps between orbits: $h\nu = E_i - E_f$.

3. Results for hydrogen-like atoms (one electron, nuclear charge Z)

Quantity	Formula	Depends on
Radius	$r_n = 0.529\,\dfrac{n^2}{Z}$ Å	$\dfrac{n^2}{Z}$
Speed	$v_n = 2.18 \times 10^6\,\dfrac{Z}{n}$ m/s	$\dfrac{Z}{n}$
Total energy	$E_n = -13.6\,\dfrac{Z^2}{n^2}$ eV	$\dfrac{Z^2}{n^2}$
Time period		$\dfrac{n^3}{Z^2}$

- Kinetic energy $= -E_n$ and potential energy $= 2E_n$. In the ground state of hydrogen: $K = 13.6$ eV, $U = -27.2$ eV, $E = -13.6$ eV.
- Hydrogen levels: -13.6, -3.4, -1.51, -0.85 eV for $n = 1, 2, 3, 4$.
- Ionisation energy from the ground state $= 13.6\,Z^2$ eV.
- Energy to excite from n_1 to n_2: $13.6\,Z^2\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right)$ eV.

4. The hydrogen spectrum

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \qquad R = 1.097 \times 10^7\ \mathrm{m}^{-1}$$

Series	Lower level n_1	Region
Lyman	1	ultraviolet

Balmer	2	visible
Paschen	3	infrared
Brackett	4	infrared
Pfund	5	infrared

- The longest wavelength of a series comes from the nearest level, $n_2 = n_1 + 1$. The shortest (the series limit) comes from $n_2 = \infty$.
- An electron falling from level n to the ground state can give at most $\dfrac{n(n-1)}{2}$ different lines.
- To turn a transition energy into a wavelength, use $\lambda(\text{nm}) = \dfrac{1240}{E(\text{eV})}$.

Solved example. Find the ratio of the longest wavelength of the Balmer series to the longest wavelength of the Lyman series.

Balmer, $3 \to 2$: $\dfrac{1}{\lambda} = R\left(\dfrac{1}{4} - \dfrac{1}{9}\right) = \dfrac{5R}{36}$. Lyman, $2 \to 1$: $\dfrac{1}{\lambda} = R\left(1 - \dfrac{1}{4}\right) = \dfrac{3R}{4}$. Ratio $= \dfrac{36/5}{4/3} = \dfrac{27}{5} = 5.4$.

Traps. The energy is negative; a higher level has a less negative energy. For ions such as He^+ and Li^{2+}, include Z^2 in energies and Z in the radius. "Second excited state" means $n = 3$. The Bohr model applies only to one-electron species.

Read in NCERT: Physics Class 12 Part 2, chapter "Atoms".

Mathematics · Matrices

Matrices and determinants together gave almost two questions per paper in 2026. Today covers matrix algebra; determinants, adjoint and inverse follow on the next two days.

1. Basics

- A matrix of order $m \times n$ has m rows, n columns and mn elements. A matrix with N elements can have as many orders as N has divisors.
- Types: row, column, square, diagonal, scalar (diagonal with equal entries), identity I, zero O, upper and lower triangular.
- The number of matrices of order $m \times n$ whose entries each come from k choices is k^{mn}.

2. Operations

- Addition needs the same order. Scalar multiplication multiplies every element.
- The product AB exists only when the number of columns of A equals the number of rows of B. If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.
- Each entry of AB is (row of A) · (column of B).
- In general $AB \neq BA$. $AB = O$ does not mean $A = O$ or $B = O$.
- Multiplication is associative and distributive: $A(BC) = (AB)C$, $A(B + C) = AB + AC$.

3. Transpose, symmetric and skew-symmetric matrices

- $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(kA)^T = kA^T$, and $(AB)^T = B^T A^T$ (the order reverses).
- Symmetric: $A^T = A$. Skew-symmetric: $A^T = -A$, so every diagonal entry is zero.
- For any square matrix, $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric, and $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.
- The trace is the sum of the diagonal entries. $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ and $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

4. Special matrices

Name	Condition
Idempotent	$A^2 = A$
Involutory	$A^2 = I$
Nilpotent	$A^k = O$ for some positive integer k
Orthogonal	$AA^T = A^T A = I$

5. Powers of a matrix

- $\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & na \\ 0 & 1 \end{pmatrix}$.
- $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$, and this matrix is orthogonal.

- Every 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies $A^2 - (a+d)A + (ad-bc)I = O$. Use this to reduce high powers.

Solved example 1. If $A^2 = A$, simplify $(I + A)^3 - 7A$.
$(I + A)^3 = I + 3A + 3A^2 + A^3$. Since $A^2 = A$, also $A^3 = A$, so this is $I + 7A$. Subtracting $7A$ leaves I.

Solved example 2. $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ satisfies $A^2 - 5A + kI = O$. Find k.
$a + d = 5$ and $ad - bc = 6 + 1 = 7$, so $A^2 - 5A + 7I = O$ and $k = 7$.

Traps. $(A + B)^2 = A^2 + AB + BA + B^2$; it equals $A^2 + 2AB + B^2$ only when $AB = BA$. Powers of $I + A$ can be expanded by the binomial theorem because I commutes with every matrix. When transposing a product, reverse the order.

Read in NCERT: Mathematics Class 12 Part 1, chapter "Matrices", with the solved examples and the miscellaneous exercise.

Chemistry · Periodicity

About one question per paper, usually asking for the correct order of a property. The general trends are easy; the marks come from knowing the exceptions.

1. The table

- Modern periodic law: properties of elements are periodic functions of their atomic numbers.
- Blocks: s (groups 1–2), p (13–18), d (3–12), f (lanthanoids and actinoids).
- **Naming elements above 100.** Digits are written as roots: 0 nil, 1 un, 2 bi, 3 tri, 4 quad, 5 pent, 6 hex, 7 sept, 8 oct, 9 enn, followed by "ium". Element 119 is ununennium; 120 is unbinilium.

2. Atomic and ionic radius

- Decreases across a period (nuclear charge rises, same shell). Increases down a group (new shells).
- A cation is smaller than its atom; an anion is larger.
- **Isoelectronic species:** the more protons, the smaller. $N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$.
- Gallium is slightly smaller than aluminium, because the ten 3d electrons shield poorly.

3. Ionisation enthalpy

- Increases across a period, decreases down a group.
- Exceptions in a period: group 13 is lower than group 2 (the electron leaves a 2p orbital, which is easier than 2s), and group 16 is lower than group 15 (half-filled p^3 is stable).
- Second period order: Li < B < Be < C < O < N < F < Ne.
- A sudden jump in successive ionisation enthalpies shows that the valence electrons have run out. A jump after the second value means a group 2 element.

4. Electron gain enthalpy

- Becomes more negative across a period. Noble gases have positive values.
- Chlorine has the most negative value of all elements. Fluorine is less negative than chlorine because the added electron enters the small, crowded 2p shell.
- Halogens: Cl > F > Br > I. Group 16: S > Se > Te > O.

5. Electronegativity and chemical character

- Electronegativity increases across a period and decreases down a group. F (4.0) > O (3.5) > N ≈ Cl (3.0).
- Metallic character decreases across a period and increases down a group.
- **Oxides** across period 3 go from basic to acidic: Na_2O and MgO are basic, Al_2O_3 is amphoteric, SiO_2, P_4O_{10}, SO_3 and Cl_2O_7 are acidic. CO, NO and N_2O are neutral.
- **Diagonal relationship:** Li resembles Mg, Be resembles Al, B resembles Si.
- Second-period elements are anomalous: small size, high electronegativity, no d orbitals, so their maximum covalency is 4.

Traps. "Electron gain enthalpy" and "electronegativity" have different top elements: chlorine for the first, fluorine for the second. In ionisation enthalpy orders, always check the Be–B and N–O pairs. For isoelectronic ions, ignore the charge sign and compare atomic numbers.

Read in NCERT: Chemistry Class 11 Part 1, chapter "Classification of Elements and Periodicity in Properties".